

SHIMONOSEKI, Japan

WMO: 477620

Lat: 33.9486N

Lon: 130.9264E

Elev: 64

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.8	36.8	17.1	13.1	38.9	19.2	14.5	40.0	26.6	40.0	24.4	40.5	12.7	300	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	90.3	78.4	88.7	78.1	87.1	77.6	80.1	86.8	79.2	85.8	78.6	85.0	7.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.3	147.2	84.2	77.4	142.7	83.7	76.5	138.7	83.2	43.8	87.0	42.9	85.9	42.2	85.3	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.6	18.9	16.5	30.5	93.1	2.8	1.7	28.5	94.3	26.8	95.3	25.3	96.2	23.2	97.4		
(5)			26.9	81.3	2.7	1.2	25.0	82.1	23.4	82.8	21.9	83.5	20.0	84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.1	44.8	45.7	50.7	58.9	66.9	73.0	80.5	83.1	77.0	68.1	58.3	49.2
	DBStd	13.78	4.29	4.74	4.88	4.72	3.62	3.45	3.88	3.03	4.14	4.30	5.08	4.95
	HDD50	430	168	134	49	2	0	0	0	0	0	0	2	75
	HDD65	2533	627	540	442	189	21	0	0	0	0	19	207	488
	CDD50	5222	7	14	72	269	525	691	945	1028	809	560	251	51
	CDD65	1850	0	0	0	6	81	242	480	563	359	113	6	0
	CDH74	14769	0	0	1	13	140	856	4591	6395	2543	229	1	0
	CDH80	4444	0	0	0	1	8	112	1425	2378	512	8	0	0
(14) Wind	WSAvg	7.0	8.8	7.9	7.8	7.3	6.8	5.8	6.1	6.1	5.7	6.0	6.8	8.5
(15) Precipitation	PrecAvg	70.0	3.1	3.4	4.5	5.4	6.2	10.8	12.5	7.3	6.9	3.3	3.7	3.0
	PrecMax	90.0	8.3	5.7	10.4	9.9	13.3	19.0	30.7	17.0	16.5	8.9	7.2	5.3
	PrecMin	43.2	1.0	1.3	2.2	1.5	1.4	2.2	2.5	1.1	2.5	0.5	1.2	0.9
	PrecStd	12.1	1.5	1.1	1.9	2.2	2.8	4.9	7.6	4.2	3.4	2.2	1.7	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.1	62.0	67.5	75.6	80.7	86.2	91.8	92.8	88.8	80.9	72.0	63.2
		MCWB	52.2	55.4	56.3	63.1	64.7	75.5	78.2	78.7	76.7	71.2	63.7	57.3
	2%	DB	54.9	57.4	63.3	71.4	77.5	82.9	89.4	91.1	86.5	78.5	69.7	60.3
		MCWB	48.7	51.5	54.6	59.8	64.0	72.9	78.3	78.8	76.3	70.0	62.7	53.7
	5%	DB	52.9	55.1	60.5	68.4	75.2	80.8	87.9	89.6	84.6	76.5	67.8	58.4
		MCWB	46.8	49.3	53.0	58.1	63.5	71.5	77.8	78.5	75.1	68.0	60.9	52.0
	10%	DB	51.2	52.8	58.2	66.1	73.0	78.5	86.3	88.1	82.9	74.7	66.1	56.5
		MCWB	45.3	46.7	51.6	57.3	63.0	69.9	77.3	78.1	74.4	66.1	59.5	50.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.2	57.1	59.7	66.0	70.5	77.3	81.0	81.6	79.4	75.0	67.2	59.6
		MCDB	56.4	60.5	64.0	70.5	74.7	83.9	87.4	88.8	85.1	78.6	70.0	61.4
	2%	WB	50.7	53.5	56.4	63.0	68.2	75.5	79.8	80.5	78.1	72.5	64.6	55.7
		MCDB	53.4	56.5	60.7	67.7	72.4	80.8	86.2	87.5	83.7	77.1	68.1	58.7
	5%	WB	48.1	50.3	54.6	61.0	66.5	74.1	78.9	79.8	77.1	70.0	62.7	53.4
		MCDB	51.6	53.6	59.2	66.0	71.3	78.4	85.4	86.6	82.6	74.8	66.3	57.0
	10%	WB	45.9	47.9	52.6	59.3	65.3	72.4	78.1	79.1	76.1	67.7	60.4	51.3
		MCDB	50.4	51.9	57.5	64.0	70.6	76.4	84.5	85.7	81.6	72.9	65.1	55.7
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	9.0	10.0	10.7	10.6	8.9	8.7	9.0	8.8	9.6	9.0	8.2
		MCDBR	10.0	12.0	13.4	14.7	13.8	11.8	10.4	10.4	9.8	10.8	10.3	10.1
	5% WB	MCWBR	7.8	8.6	8.9	7.7	6.6	5.2	4.0	4.0	4.5	5.8	7.1	7.8
		MCWBR	9.5	10.2	11.3	11.4	10.4	9.3	9.5	9.4	8.1	8.6	8.8	8.8
(40) Clear-Sky Solar Irradiance	taub		0.429	0.470	0.533	0.555	0.559	0.574	0.572	0.534	0.502	0.457	0.431	0.421
		taud	2.060	1.987	1.839	1.840	1.890	1.940	1.992	2.098	2.137	2.160	2.145	2.095
	Ebn at Noon		236	241	236	239	239	235	234	241	243	243	235	230
		Edh at Noon	42	50	63	65	63	60	56	50	46	42	38	38
(44) All-Sky Solar Radiation	RadAvg		680	910	1255	1527	1669	1441	1554	1629	1292	1091	806	632
	RadStd		61	89	91	130	190	157	235	180	131	113	74	42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page